



Database Management System Project

[Topic : Queries]

COLLEGE FESTIVAL MANAGEMENT SYSTEM

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Query 01

-- Question : List all festivals with their organizing college, committee, duration in days, and festival type – ordered by start date.

```
SELECT F.Festival_ID,

       F.Festival_Name,

       F.Festival_Type,

       F.Start_Date,

       F.End_Date,

       (JULIANDAY(F.End_Date) - JULIANDAY(F.Start_Date) + 1) AS
Duration_Days,

       C.Committee_Name,

       CL.College_Name

FROM Festival F

JOIN Committee C  ON F.Committee_ID = C.Committee_ID

JOIN College   CL ON C.College_ID   = CL.College_ID

ORDER BY F.Start_Date;
```

	festival_id integer	festival_name character varying (150)	festival_type character varying (50)	start_date date	end_date date	duration_days integer	committee_name character varying (150)	college_name character varying (150)
1	3	iCultura 2025	Cultural	2025-01-20	2025-01-22	3	Fest Core Committee	DAIICT
2	1	Nirmaitsav 2025	Cultural	2025-02-14	2025-02-16	3	Cultural Committee	Nirma University
3	6	SportzMania 2025	Sports	2025-02-20	2025-02-22	3	Sports Committee	CHARUSAT University
4	2	TechFest 2025	Technical	2025-03-01	2025-03-03	3	Technical Committee	Nirma University
5	5	Aaveg 2025	Cultural	2025-03-15	2025-03-17	3	Management Committee	Ahmedabad University
6	4	Synergy 2025	Mixed	2025-04-05	2025-04-07	3	Events Committee	LD College of Engineeri...

```
-- =====
-- Query 02
-- Question : Full festival dashboard – events, unique participants,
-- stalls, and total revenue (completed payments only) per festival.
-- Features : 5-table LEFT JOIN + CASE + GROUP BY (dashboard query)
-- =====

SELECT F.Festival_Name,
       F.Festival_Type,
       F.Start_Date,
       F.End_Date,
       COUNT(DISTINCT E.Event_ID)
AS Total_Events,
       COUNT(DISTINCT R.Person_ID)
AS Total_Participants,
       COUNT(DISTINCT S.Stall_ID)
AS Total_Stalls,
       COALESCE(SUM(CASE WHEN PAY.Payment_Status = 'Completed'
                        THEN PAY.Payment_Amount END), 0)
AS Total_Revenue
FROM Festival F
LEFT JOIN Event      E      ON F.Festival_ID = E.Festival_ID
LEFT JOIN Registration R    ON E.Event_ID    = R.Event_ID
LEFT JOIN Stall      S      ON F.Festival_ID = S.Festival_ID
LEFT JOIN Ticket      T      ON E.Event_ID    = T.Event_ID
LEFT JOIN Payment     PAY    ON T.Ticket_ID    = PAY.Ticket_ID
GROUP BY F.Festival_ID, F.Festival_Name, F.Festival_Type,
         F.Start_Date, F.End_Date
ORDER BY Total_Revenue DESC;
```

	festival_name character varying (150) 🔒	festival_type character varying (50) 🔒	start_date date 🔒	end_date date 🔒	total_events bigint 🔒	total_participants bigint 🔒	total_stalls bigint 🔒	total_revenue numeric 🔒
1	Nirmaltsav 2025	Cultural	2025-02-14	2025-02-16	4	8	3	8250.00
2	TechFest 2025	Technical	2025-03-01	2025-03-03	3	6	3	6600.00
3	Aaveg 2025	Cultural	2025-03-15	2025-03-17	2	4	2	3320.00
4	iCultura 2025	Cultural	2025-01-20	2025-01-22	2	5	2	1680.00
5	Synergy 2025	Mixed	2025-04-05	2025-04-07	2	4	2	960.00
6	SportzMania 2025	Sports	2025-02-20	2025-02-22	2	3	2	560.00

```
-- =====
-- Query 03
-- Question : Find participants who registered for more than one event -
-- show name, college, and event count.
-- Features : GROUP BY + HAVING (filter after aggregation)
-- =====

SELECT P.Person_ID,
       P.Name,
       P.Gender,
       P.College_Name,
       COUNT(R.Event_ID) AS Events_Registered
FROM Participant P
JOIN Registration R ON P.Person_ID = R.Person_ID
GROUP BY P.Person_ID, P.Name, P.Gender, P.College_Name
HAVING COUNT(R.Event_ID) > 1
ORDER BY Events_Registered DESC;
```

	person_id [PK] integer	name character varying (100)	gender character varying (10)	college_name character varying (150)	events_registered bigint
1	10	Nidhi Bhatt	Female	Ahmedabad University	2
2	6	Mahi Desai	Female	Nirma University	2
3	2	Diya Shah	Female	DAIICT	2
4	9	Harsh Patel	Male	Nirma University	2
5	7	Vivek Rao	Male	DAIICT	2
6	12	Krishna Kapoor	Female	Nirma University	2
7	3	Rohan Mehta	Male	LD College of Engineeri...	2
8	1	Aarav Patel	Male	Nirma University	2
9	5	Kush Trivedi	Male	CHARUSAT University	2
10	8	Anvi Sharma	Female	LD College of Engineeri...	2

```
-- =====
-- Query 04
-- Question : Rank all events by revenue using a window function (RANK
OVER). Show rank, event name, festival, and total revenue.
-- Features : Subquery in FROM + RANK() window function
-- =====

SELECT RANK() OVER (ORDER BY COALESCE(REV.Revenue, 0) DESC) AS
Revenue_Rank,
    E.Event_Name,
    E.Event_Category,
    F.Festival_Name,
    E.Ticket_Price,
    COALESCE(REV.Revenue, 0) AS Total_Revenue
FROM Event E
JOIN Festival F ON E.Festival_ID = F.Festival_ID
LEFT JOIN (
    SELECT T.Event_ID,
        SUM(PAY.Payment_Amount) AS Revenue
    FROM Ticket T
    JOIN Payment PAY ON T.Ticket_ID = PAY.Ticket_ID
    WHERE PAY.Payment_Status = 'Completed'
    GROUP BY T.Event_ID
) REV ON E.Event_ID = REV.Event_ID
ORDER BY Revenue_Rank;
```

	revenue_rank bigint	event_name character varying (150)	event_category character varying (100)	festival_name character varying (150)	ticket_price numeric (10,2)	total_revenue numeric
1	1	DJ Night	Music	Nirmaitsav 2025	300.00	600.00
2	2	Stand Up Comedy	Comedy	Aaveg 2025	180.00	540.00
3	3	Robo Wars	Robotics	TechFest 2025	250.00	500.00
4	4	Hackathon 2025	Technical	TechFest 2025	200.00	400.00
5	5	Dance Fever	Dance	Nirmaitsav 2025	100.00	300.00
6	5	Battle of Bands	Music	Nirmaitsav 2025	150.00	300.00
7	7	Fashion Show	Fashion	Synergy 2025	120.00	240.00
8	7	Singing Competition	Music	iCultura 2025	80.00	240.00
9	9	Cricket Tournament	Sports	SportzMania 2025	70.00	140.00
10	10	Treasure Hunt	Adventure	iCultura 2025	60.00	60.00
11	11	Nukkad Natak	Drama	Nirmaitsav 2025	50.00	50.00
12	12	Photography Contest	Art	Aaveg 2025	40.00	40.00
13	13	Quiz Competition	Academic	Synergy 2025	30.00	0
14	13	Paper Presentation	Academic	TechFest 2025	0.00	0.00
15	13	Badminton Champions...	Sports	SportzMania 2025	50.00	0

```
-- =====
-- Query 05
-- Question : Show payment status breakdown – count and total amount for
each status (Completed, Pending, Failed, Refunded).
-- Features : COUNT/SUM + SUM() OVER() window (percentage)
-- =====

SELECT PAY.Payment_Status,
       COUNT(PAY.Payment_ID)      AS Transaction_Count,
       SUM(PAY.Payment_Amount)    AS Total_Amount,
       ROUND(100.0 * COUNT(PAY.Payment_ID) /
             SUM(COUNT(PAY.Payment_ID)) OVER (), 2) AS Percentage
FROM Payment PAY
GROUP BY PAY.Payment_Status
ORDER BY Transaction_Count DESC;
```

	payment_status character varying (20) 🔒	transaction_count bigint 🔒	total_amount numeric 🔒	percentage numeric 🔒
1	Completed	25	3410.00	83.33
2	Pending	2	260.00	6.67
3	Failed	2	60.00	6.67
4	Refunded	1	50.00	3.33

```

-- =====
-- Query 06
-- Question : Find participants who registered for events but have NOT
made any payment – using NOT EXISTS.
-- Features : NOT EXISTS correlated subquery
-- =====

SELECT DISTINCT
    P.Person_ID,
    P.Name,
    P.Email,
    P.College_Name,
    E.Event_Name,
    R.Registration_Date
FROM Participant P
JOIN Registration R ON P.Person_ID = R.Person_ID
JOIN Event E ON R.Event_ID = E.Event_ID
WHERE NOT EXISTS (
    SELECT 1
    FROM Payment PAY
    JOIN Ticket T ON PAY.Ticket_ID = T.Ticket_ID
    WHERE PAY.Person_ID = P.Person_ID
        AND T.Event_ID = E.Event_ID
)
ORDER BY P.Name;

```

	person_id integer	name character varying (100)	email character varying (100)	college_name character varying (150)	event_name character varying (150)	registration_date date
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```
-- =====
-- Query 07
-- Question : List each organizer with the number of sponsors they manage
and total sponsorship amount – ordered by total managed.
-- Features : Multi-table JOIN + LEFT JOIN + GROUP BY + COALESCE
-- =====

SELECT O.Organizer_ID,
       O.Organizer_Name,
       O.Department,
       C.Committee_Name,
       CL.College_Name,
       COUNT(SP.Sponsor_ID)           AS Sponsors_Managed,
       COALESCE(SUM(SP.Sponsorship_Amount), 0) AS Total_Sponsorship
FROM Organizer O
JOIN Committee C      ON O.Committee_ID = C.Committee_ID
JOIN College   CL     ON C.College_ID   = CL.College_ID
LEFT JOIN Sponsor SP  ON O.Organizer_ID = SP.Organizer_ID
GROUP BY O.Organizer_ID, O.Organizer_Name, O.Department,
         C.Committee_Name, CL.College_Name
ORDER BY Total_Sponsorship DESC;
```

	organizer_id integer	organizer_name character varying (100)	department character varying (100)	committee_name character varying (150)	college_name character varying (150)	sponsors_managed bigint	total_sponsorship numeric
1	4	Karan Desai	Mechanical	Events Committee	LD College of Engineeri...	2	105000.00
2	5	Sneha Joshi	Civil	Management Committee	Ahmedabad University	2	90000.00
3	1	Riya Shah	Computer Science	Cultural Committee	Nirma University	2	75000.00
4	2	Arjun Mehta	Electronics	Technical Committee	Nirma University	2	70000.00
5	3	Pooja Patel	IT	Fest Core Committee	DAIICT	2	35000.00
6	6	Dev Trivedi	Computer Science	Cultural Committee	Nirma University	0	0
7	8	Mihir Shah	Electrical	Sports Committee	CHARUSAT University	0	0
8	7	Ananya Rao	MBA	Fest Core Committee	DAIICT	0	0


```
-- =====
-- Query 08
-- Question : Show college-wise participation summary – total unique
participants and total money spent across all festivals.
-- Features : LEFT JOIN + CASE + GROUP BY across 3 tables
-- =====

SELECT P.College_Name,
       COUNT(DISTINCT P.Person_ID) AS Total_Participants,
       COUNT(R.Registration_ID) AS Total_Registrations,
       COALESCE(SUM(CASE WHEN PAY.Payment_Status = 'Completed'
                        THEN PAY.Payment_Amount END), 0) AS Total_Spent
FROM Participant P
LEFT JOIN Registration R ON P.Person_ID = R.Person_ID
LEFT JOIN Payment PAY ON P.Person_ID = PAY.Person_ID
GROUP BY P.College_Name
ORDER BY Total_Spent DESC;
```

	college_name character varying (150) 🔒	total_participants bigint 🔒	total_registrations bigint 🔒	total_spent numeric 🔒
1	Nirma University	5	17	3350.00
2	LD College of Engineeri...	4	10	1000.00
3	DAIICT	4	10	720.00
4	CHARUSAT University	3	6	650.00
5	Ahmedabad University	4	7	380.00

```
-- =====
-- Query 09
-- Question : Find the most expensive event in each festival using a
correlated subquery.
-- Features : Correlated subquery in WHERE
-- =====

SELECT F.Festival_Name,
       E.Event_Name,
       E.Event_Category,
       E.Event_Venue,
       E.Ticket_Price AS Max_Ticket_Price
FROM Event E
JOIN Festival F ON E.Festival_ID = F.Festival_ID
WHERE E.Ticket_Price = (
    SELECT MAX(E2.Ticket_Price)
    FROM Event E2
    WHERE E2.Festival_ID = E.Festival_ID
)
ORDER BY Max_Ticket_Price DESC;
```

	festival_name character varying (150) 🔒	event_name character varying (150) 🔒	event_category character varying (100) 🔒	event_venue character varying (200) 🔒	max_ticket_price numeric (10,2) 🔒
1	Nirmaitsav 2025	DJ Night	Music	Open Ground	300.00
2	TechFest 2025	Robo Wars	Robotics	Sports Ground	250.00
3	Aaveg 2025	Stand Up Comedy	Comedy	Main Auditorium	180.00
4	Synergy 2025	Fashion Show	Fashion	Ramp Area	120.00
5	iCultura 2025	Singing Competition	Music	Auditorium	80.00
6	SportzMania 2025	Cricket Tournament	Sports	Cricket Ground	70.00

```

-- =====
-- Query 10
-- Question : List all volunteers with the number of events assigned to
-- them – highlight volunteers not yet assigned to any event.
-- Features : LEFT JOIN + CASE expression + GROUP BY
-- =====

SELECT V.Volunteer_ID,
       V.Volunteer_Name,
       V.Volunteer_Role,
       COUNT(VA.Event_ID)                                AS Events_Assigned,
       CASE WHEN COUNT(VA.Event_ID) = 0
            THEN 'Not Assigned'
            ELSE 'Active'
       END                                                AS Status
FROM Volunteer V
LEFT JOIN Volunteer_Assignment VA ON V.Volunteer_ID = VA.Volunteer_ID
GROUP BY V.Volunteer_ID, V.Volunteer_Name, V.Volunteer_Role
ORDER BY Events_Assigned DESC;

```

	volunteer_id [PK] integer	volunteer_name character varying (100)	volunteer_role character varying (100)	events_assigned bigint	status text
1	1	Neel Shah	Stage Manager	3	Active
2	3	Parth Mehta	Security	2	Active
3	5	Dhruv Joshi	Logistics	2	Active
4	4	Ritu Sharma	Photography	2	Active
5	6	Kavya Desai	Event Coordinator	2	Active
6	2	Isha Patel	Registration Desk	2	Active
7	7	Manan Trivedi	Technical Support	2	Active
8	8	Shreya Rao	Hospitality	2	Active
9	9	Aditya Kumar	Crowd Management	2	Active
10	10	Puja Singh	First Aid	1	Active

```
-- =====
-- Query 11
-- Question : Show month-wise registration trend – total registrations and
unique participants per month.
-- Features : STRFTIME date function + GROUP BY + COUNT DISTINCT
-- =====

SELECT STRFTIME('%Y-%m', R.Registration_Date) AS Month,
       COUNT(R.Registration_ID)      AS Total_Registrations,
       COUNT(DISTINCT R.Person_ID)   AS Unique_Participants
FROM Registration R
GROUP BY STRFTIME('%Y-%m', R.Registration_Date)
ORDER BY Month;
```

	month 	total_registrations 	unique_participants 
1	2025-01	5	5
2	2025-02	17	15
3	2025-03	8	8

```
-- =====
-- Query 12
-- Question : For each festival, show merchandise inventory summary –
-- number of item types, total stock, and total inventory value.
-- Features : JOIN + aggregate functions (MIN, MAX, SUM, COUNT)
-- =====

SELECT F.Festival_Name,
       COUNT(M.Merchandise_ID)           AS Item_Types,
       SUM(M.Stock_Quantity)              AS Total_Stock,
       SUM(M.Price * M.Stock_Quantity)    AS Total_Inventory_Value,
       MIN(M.Price)                       AS Cheapest_Item,
       MAX(M.Price)                       AS Most_Expensive_Item
FROM Festival F
JOIN Merchandise M ON F.Festival_ID = M.Festival_ID
GROUP BY F.Festival_ID, F.Festival_Name
ORDER BY Total_Inventory_Value DESC;
```

	festival_name character varying (150) 🔒	item_types bigint 🔒	total_stock bigint 🔒	total_inventory_value numeric 🔒	cheapest_item numeric 🔒	most_expensive_item numeric 🔒
1	Nirmaitsav 2025	4	850	161650.00	49.00	599.00
2	TechFest 2025	3	460	96540.00	99.00	399.00
3	Synergy 2025	2	340	69160.00	79.00	549.00
4	Aaveg 2025	2	360	56440.00	59.00	279.00
5	SportzMania 2025	2	195	44055.00	149.00	349.00
6	iCultura 2025	2	420	38580.00	49.00	199.00

```
-- =====
-- Query 13
-- Question : Find events where the number of volunteers assigned is
greater than or equal to the number of participants registered.
-- Features : LEFT JOIN + GROUP BY + HAVING on two COUNT DISTINCTs
-- =====

SELECT E.Event_ID,
       E.Event_Name,
       F.Festival_Name,
       COUNT(DISTINCT VA.Volunteer_ID) AS Volunteers,
       COUNT(DISTINCT R.Person_ID)     AS Participants
FROM Event E
JOIN Festival F ON E.Festival_ID = F.Festival_ID
LEFT JOIN Volunteer_Assignment VA ON E.Event_ID = VA.Event_ID
LEFT JOIN Registration R ON E.Event_ID = R.Event_ID
GROUP BY E.Event_ID, E.Event_Name, F.Festival_Name
HAVING COUNT(DISTINCT VA.Volunteer_ID) >= COUNT(DISTINCT R.Person_ID)
ORDER BY Volunteers DESC;
```

	event_id integer	event_name character varying (150)	festival_name character varying (150)	volunteers bigint	participants bigint
1	5	Robo Wars	TechFest 2025	2	2
2	1	Battle of Bands	Nirmaitsav 2025	2	2
3	15	DJ Night	Nirmaitsav 2025	2	2
4	14	Badminton Champions...	SportzMania 2025	1	1
5	6	Paper Presentation	TechFest 2025	1	1
6	3	Nukkad Natak	Nirmaitsav 2025	1	1
7	12	Photography Contest	Aaveg 2025	1	1

```
-- =====
```

```
-- Query 14
-- Question : Using a CTE, find the top 5 participants by total amount
paid across all events (completed payments only).
-- Features : CTE (WITH clause) + RANK() window function + LIMIT
-- =====

WITH ParticipantSpend AS (
    SELECT P.Person_ID,
           P.Name,
           P.Gender,
           P.College_Name,
           SUM(PAY.Payment_Amount) AS Total_Paid,
           COUNT(PAY.Payment_ID)   AS Payment_Count
    FROM Participant P
    JOIN Payment PAY ON P.Person_ID = PAY.Person_ID
    WHERE PAY.Payment_Status = 'Completed'
    GROUP BY P.Person_ID, P.Name, P.Gender, P.College_Name
)
SELECT RANK() OVER (ORDER BY Total_Paid DESC) AS Rank,
       Person_ID,
       Name,
       Gender,
       College_Name,
       Payment_Count,
       Total_Paid
FROM ParticipantSpend
ORDER BY Total_Paid DESC
LIMIT 5;
```

	rank bigint 🔒	person_id [PK] integer 🔒	name character varying (100) 🔒	gender character varying (10) 🔒	college_name character varying (150) 🔒	payment_count bigint 🔒	total_paid numeric 🔒
1	1	9	Harsh Patel	Male	Nirma University	2	550.00
2	2	12	Krishna Kapoor	Female	Nirma University	2	380.00
3	2	6	Mahi Desai	Female	Nirma University	2	380.00
4	4	1	Aarav Patel	Male	Nirma University	2	330.00
5	5	5	Kush Trivedi	Male	CHARUSAT University	2	250.00